

An Emotion-Aware Voice Agent with Mood Tracking and Emotional Memory

1. V Shanmukha Sathwik Atchutuni 2. Soumith Rao Amradi 3. Nithin Yadav Godetti 4. Sidhartha Reddy Ravula | Dr.Clifford Whitworth | The Anuradha and Vikas Sinha Department of Data Science

INTRODUCTION / BACKGROUND

- **Problem.** Despite advances in speech emotion recognition [1], most conversational agents treat emotion as a one-off, per-turn signal rather than a dynamic state, so empathy is reactive, not adaptive across the conversation. Emotional meaning also lives in *how* we speak, not just words, a limit of text-only systems [2].
- **Compounding gap.** Long-term memory agents store user facts without emotional context, so what they recall is not weighted by what mattered to the user [3].
- **Our method.** We combine a decaying MoodManager (response tone) with an emotion-weighted memory (response content) into a single voice-based agent.

RESEARCH QUESTION / OBJECTIVES

Research Questions:

- **RQ1:** Does a decaying mood state produce smoother, more consistent tone shifts than reacting to each turn's emotion in isolation?
- **RQ2:** Does emotion-weighted long-term memory improve how personalized and understood users feel, compared to an agent with no memory?

Objectives

- Build the end-to-end pipeline: Whisper ASR + Bi-LSTM emotion recognition -> MoodManager -> RAG memory -> LLM reply + expressive TTS
- Evaluate each layer: emotion classifier (macro-F1), mood smoothing (ablation), memory recall (two-session test), user study (ON vs OFF)

METHODS

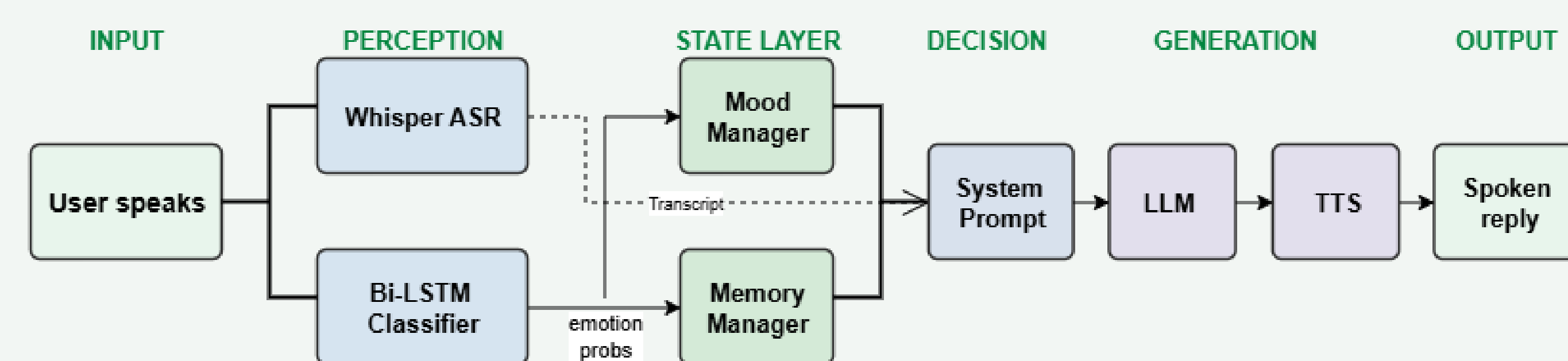


Fig. 1. System Architecture

- **Data & Models.** RAVDESS + CREMA-D (planned). Features (MFCC, F0, energy, spectral) -> RF/XGBoost -> CNN -> Bi-LSTM.
- **Dual State Layer.** *MoodManager* -> decaying tone (**how** it speaks). *MemoryManager* -> emotion-weighted RAG, salience gates storage (**what** it recalls).

RESULTS / FINDINGS

Strongest result from the Phase-1 perception baseline.

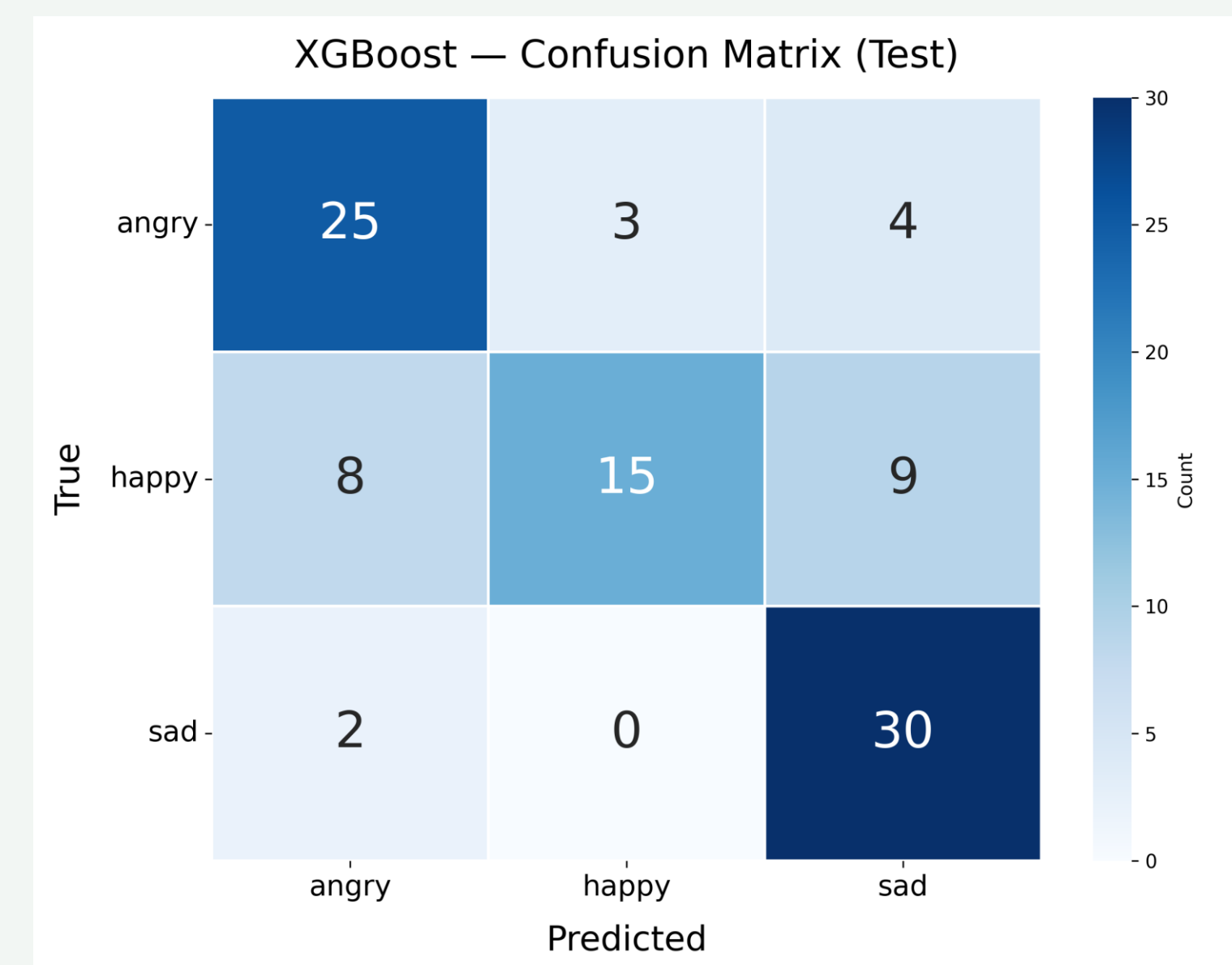


Fig. 2. XGBoost Confusion Matrix, 3-Class Emotion Recognition

Caption: Sad 30/32 and angry 25/32 are strong; happy is weakest at 15/32, split toward angry (8) and sad (9).

Takeaway: High- and low-arousal emotions separate cleanly; happy is confused both ways, the gap that motivates temporal modeling next.

RESULTS / ANALYSIS

Model	Accuracy	Macro-f1
Random Forest	0.667	0.657
XGBoost	0.729	0.715

Table 1. Classical Baseline Comparison

- Performance reported from XGBoost; feature attribution from Random Forest, cleaner, more interpretable importance scores.

- XGBoost is the stronger baseline; arousal separates well (sad, angry), while happy is confused both ways.
- Energy drives the decision, per-clip stats miss the emotional contour, motivating the **Bi-LSTM**.

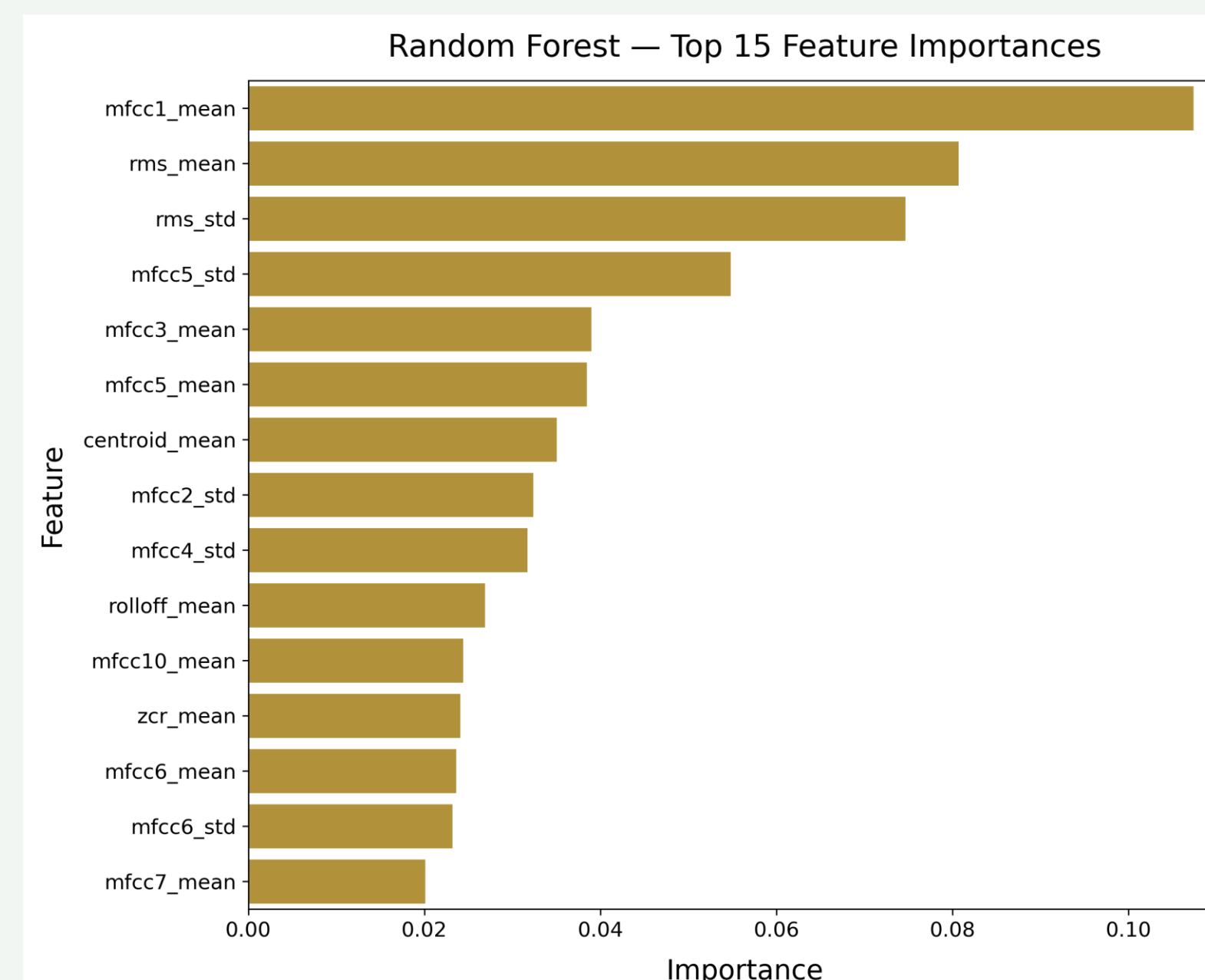


Fig. 3. MFCCs and RMS energy dominate; pitch (F0) falls outside the top 15

DISCUSSION

- Classical baselines confirm the feature pipeline is sound, sad and angry are acoustically well-separated
- Happy overlaps with both classes, revealing the limits of static feature vectors and motivating temporal sequence modeling
- Bi-LSTM is designed to capture prosodic patterns over time that a single summary vector misses
- Current results are RAVDESS only, CREMA-D integration planned for broader speaker-independent generalization
- Validated perception layer directly feeds MoodManager and MemoryManager, the two state layers driving tone and content in the full pipeline

CONCLUSIONS / NEXT STEPS

- Phase 1 validates the acoustic feature pipeline and sets a strong classical baseline for 3-class emotion recognition.
- The dual memory design, a decaying MoodManager for tone and an emotion-gated MemoryManager for facts, both driven by one speech-derived emotion signal, is the core contribution.
- The goal is a personal voice companion that reads emotion, remembers the user, and responds like a close friend, built to make people feel heard, not to act as a professional tool

REFERENCES

- [1] R. A. Khalil *et al.*, "Speech Emotion Recognition Using Deep Learning: A Review," *IEEE Access*, vol. 7, pp. 117327–117345, 2019.
- [2] L. Wang *et al.*, "Empathetic Dialogue Generation via Sensitive Emotion Recognition and Sensible Knowledge Selection," *Findings of ACL: EMNLP*, 2022.
- [3] X. Xu *et al.*, "Long Time No See! Open-Domain Conversation with Long-Term Persona Memory," *Findings of ACL*, 2022.

ACKNOWLEDGMENTS / CONTACT

We thank Dr. Clifford Whitworth (College of Information, University of North Texas), Laxmigayathri Challa, and Shiny Shamma Kota for mentorship on this project, developed for DTSC 5082. Emotion data: RAVDESS corpus.